

4 (a) (i) $V_{out} = 0.040 \times 80 = 3.2\text{V}$ [M1]
 potential at A= 0-3.2V=-3.2V[A1]

(ii) $V_x = 8.0 - 3.2 = 4.8\text{V}$ [C1]
 $R_x = \frac{4.8}{0.040} = 120\Omega$ [A1]

(b) (i) $R_{out} = \frac{100 \times 80}{100 + 80} = 44.44\Omega$ [C1]
 $V_{out} = \frac{R_{out}}{R_{total}} \times 8.0$
 $= \frac{44.44}{44.44 + 120} \times 8.0$ [C1]
 $= 2.16\text{ V}$ [A1]



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- (ii) When temperature gets hotter, resistance across the thermistor decreases [M1]

and hence effective/combined resistance across AB decreases. By applying potential divider rule, potential difference across BC increases [M1] which will turn on the fan.

Thus, fan should be connected across BC [A1]

- 5 (a) (i) The magnetic flux density of a magnetic field is defined as the force per unit